Array-9

26 November 2025 20:05

Search in a rotated sorted array

{3, 4, 5, 6, 7, 0, 1, 2} , Target = 4

Initial

$l = 0, r = n-1 \Rightarrow 8-1 = 7$

while ($l < r$)

$mid = l + (r-l)/2$

3	4	5	6	7	0	1	2
l			m				r

$mid = 0 + \frac{(7-0)}{2} \Rightarrow 3.5$

$[l] < [m] \Rightarrow 3 < 6$

$[l] < Target$ && $Target < [m]$

$3 < 4$ && $4 < 6$

So $r = mid - 1 \Rightarrow 3 - 1 = 2$

3	4	5	6	7	0	1	2
l	m	r					

$mid = 0 + \frac{(2-0)}{2} \Rightarrow 1$

$[m] = Target = 4$

Return 4

Peak element

{3, 4, 5, 6, 7, 0, 1, 2}

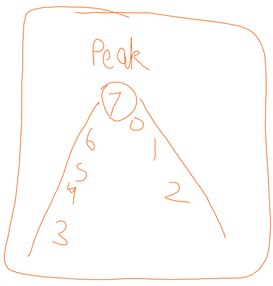
$l = 0, r = 7$

while ($l < r$)

$m = l + (r-l)/2$

$m = 3$

3	4	5	6	7	0	1	2
l			m				r



$[m] < [m+1] \Rightarrow 6 < 7$

$l = m+1$

$= 3+1 = 4$

3	4	5	6	7	0	1	2
				l	m		r

$m = 4 + \frac{(7-4)}{2} \Rightarrow 4 + 1 \Rightarrow 5$

$[m] < [m+1] \Rightarrow 0 < 1$

$\Rightarrow l = m+1 \Rightarrow 6$

3	4	5	6	7	0	1	2
						l m	r

$m = 6 + \frac{(7-6)}{2} \Rightarrow 6 + 0 \Rightarrow 6$

So $[m] < [mid+1]$

$1 < 2$

$l = m+1 = 7$

loop ends

Return 2